

ALASSANE DIALLO

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3 Rue justice pourpre, Cergy, France

EDUCATION

University PARIS Diderot, Paris, France

Master of Science in Financial Engineering (M2MO), Distinction -| 2020-2021

Relevant Coursework: Fixed Income, Quantitative Analysis, Machine learning for finance, Financial Risk Modeling. Partial Differential Equations, Stochastic Modeling, Financial Engineering, Reinforcement learning, Natural language Processing.

ECOLE CENTRALE-PARIS Paris, France

Master's Degree in Applied Mathematics, High Honor -| 2017-2021

Relevant courses: Advanced Statistics, Deep learning, Reinforcement learning, Portfolio Optimization, Algorithmic trading, Natural Language Processing, Bayesian Statistics

GRENOBLE ALPES UNIVERSITY

Bachelor of Mathematics and Physics Top 1% (GP A: 4.0/4.0) -| 2014-2017

- Awarded the National Merit Scholarship by the Malian Ministry of Higher Education and Scientific Research.
- **Relevant Course:** Statistical Quantum Physics, C++, Mechanics, Computer Science for Physics, Electromagnetism.

COMPETENCES

Programming languages: C++, Python, VBA, Latex, Databases: MYSQL

Control System: Git/Github, Software Tools: Turtoise, Postman, Eikon

Office: Excel, Word, Powerpoint, Languages Spoken: French (Mother tongue), English (Fluent).

Knowledge Financial Market:

Pricing of interest rate derivatives: swap, cap, floor

Knowledge of Forex: spot, forward, swap.

Pricing of options and structured product

PROFESSIONAL EXPERIENCES

Udemy platform

Professor of Mathematics, Computer Science, and Physics -| 2023 - Present

Teaches financial markets, risk management, and artificial intelligence at the master's level on Udemy.

I have published the following courses:

- Deep reinforcement learning with Python
- Machine learning with Python
- Risk management with Python
- Credit risk management with Python
- Enterprise risk management with Python
- Hedge fund modelling with Python
- Stochastic modeling with Python
- Bayesian statistics with Python
- Mathematical modeling with Excel
- Time series analysis with Python and Excel

Skills: Financial Education, Artificial Intelligence Instruction, Risk Management Courses, Python Programming

Superprof and MeetinClass, France, Paris :

Professor of Mathematics -Computer Science and Physics -| 2019- Present

- Tutoring and support for mathematics, physics, and computer science students.
- Offering personalized learning experiences tailored to individual student needs and learning styles.
- Superprof award for outstanding performance and contributions to student success.
- Effective teaching methodologies to enhance student understanding.
- Staying current with curriculum changes, educational trends, and advancements in mathematics, physics, and computer science to deliver up-to-date and relevant instruction.

Skills: Personalized Learning, Effective Teaching Methodologies, Student Success, Curriculum Innovation

LONDON STOCK EXCHANGE GROUP, Paris, France

Quantitative Analyst -| 2021-Present

- In charge of the validation of rate/forex models of the QPS (Quantitative Pricing Service) pricer within the Refinitiv model validation team, and the development/maintenance in Python of the pricing library of the validation team of the model.
- Validation of interest rate products (ILB, FRB, FRN, FRFF).
- Validation of FX structured products (FX TARN, FX TARF, FX-swap, FX-option).
- Extension of the pricing library of the model validation team in Python and development of python program for mass testing automation. using multithreading method and asynchronous programming.

Skills: Python, Postman, Tortoise, Excel, Git/GitHub, Forex, and Interest rate Markets.

QUANTEAM, Paris, France

Quantitative Analyst -| 2021-Present

- Working on Deep reinforcement learning algorithms and Bayesian deep reinforcement learning methods for market making. and portfolio Optimization within the Quant Practice Team.
- Contribute to the writing of Quant documents and articles.

Skills: Python, Paper readings, C++, PyTorch, Latex, Git/Github, AI research, Portfolio Optimization.

ESTIMANCY, Paris, France

Deep learning/NLP intern -| 2021-2022

- Propose NLP methods based on Deep Learning for measuring user requirements
- Perform bibliographic research on Deep Learning methods used for entity extraction (NER). Studied and implemented different algorithms including CNN (Convolutional Neural Network) for entity extraction, LSTM.
- RNN (Recurrent Neural Network), BI-LSTM (Bidirectional long, short-term memory), CRF (conditional random field), BERT

Skills: Deep learning, NLP, Python, Pytorch, AI research, Paper implementations.

MAZARS, Paris, France

Quantitative Researcher -| 2020-21

Implementation of a credit risk management model called JLT for enriching the pricing library

- Reading research articles on credit risk around the JLT model
- Implementation calibration, and Validation of the JLT Model on Python
- Study of model sensitivities by variation of risk premium parameters Technical environment: Python.

Skills: Python, Credit risk modelling, team work.

LPSM, Paris, France

Quantitative Analyst/AI Intern -| May 2020-Sep-2020

- Development of Deep Reinforcement Learning Model for Financial Assets Portfolio Optimization
- Tasks involved: Reading research articles on artificial intelligence applied to portfolio optimization.
- Study of deep learning / Reinforcement learning algorithms including Deep Deterministic policy gradient and Deep Q network

Skills: Python, Deep Reinforcement Learning, Portfolio optimization, PyTorch.